T3 Firmware & Bootloader

The update registers · built Nov 13, 2026

The update registers

The host drives the whole update through a handful of Modbus registers: the control registers that start and stop it, the checksum registers that decide whether the image is the intended one, and the status register that says where the update has got to.

3. Modbus Register Mapping for Firmware Update

Update Control Registers

Register	Address	Type	Purpose	Values
UPDATE_STATUS	16	W	Firmware update trigger	See below
UPDATE_PTR_HI	17	R/W	Resume position HIGH byte	0x00-0xFF
UPDATE_PTR_LO	18	R/W	Resume position LOW byte	0x00-0xFF
UPDATE_FLAG	23	R/W	Update completion status	0x55 = success
ISP_RESUME_COUNT	1991	R/W	Packet count for resume	0-65535

MD5 Checksum Registers

Register	Address	Type	Purpose	Size
MD5[0]	1995	R/W	MD5 checksum word 0	16-bit
MD5[1]	1996	R/W	MD5 checksum word 1	16-bit
MD5[2]	1997	R/W	MD5 checksum word 2	16-bit
MD5[3]	1998	R/W	MD5 checksum word 3	16-bit

Test Buffer Registers (Diagnostic)

Register	Address	Range	Purpose
TEST_BUF	100-150	100-150	Diagnostic counters

4. UPDATE_STATUS Register (Address 16) - Command Values

TEXT

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UPDATE_STATUS write values trigger different firmware update modes:


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│ Write Value │ Command                  │ Action              │
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│ 0x00        │ PROGRAM_FINISH           │ End OTA update      │
│ 0x01        │ PROGRAM_FINISH           │ End OTA update      │
│ 0x1F        │ PROGRAM_RESUME           │ Start/Resume update │
│ 0x3F        │ ERASE_ALL_FLASH          │ Erase OTA partition │
│ 0x7F        │ PROGRAM_START (unused)   │ Reserved            │
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Detailed Command Descriptions:


1. Write 0x1F to UPDATE_STATUS (Address 16):
   ├─ Start new firmware update OR resume interrupted update
   ├─ Retrieves resume position from flash (FLASH_RS_RES_PACKAGE)
   ├─ Calls esp_ota_begin() to prepare OTA partition
   ├─ Expects subsequent firmware data on UPDATE_PTR registers
   └─ Updates com_flash_high/low for resume tracking

2. Write 0x01 to UPDATE_STATUS (Address 16):
   ├─ Complete firmware update process
   ├─ Call esp_ota_end() to finalize image
   ├─ Validate image checksum
   ├─ Call esp_ota_set_boot_partition() to mark active
   ├─ Set UPDATE_FLAG = 0x55 on success
   ├─ Clear resume counter
   └─ Device ready to restart

3. Write 0x3F to UPDATE_STATUS (Address 16):
   ├─ Full erase of OTA partition
   ├─ Reset all resume tracking data
   ├─ Clear flash resume counter
   ├─ Abort any in-progress update
   ├─ com_update_firmware = 0
   └─ Ready for new update from packet 0
```